

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$\begin{matrix} 8x + 7y = 9 \\ 24x + 21y = 27 \end{matrix}$$

For each real number r , which of the following points lies on the graph of each equation in the xy -plane for the given system?

Answer

- A. $(r, -\frac{8r}{7} + \frac{9}{7})$
- B. $(-\frac{8r}{7} + \frac{9}{7}, r)$
- C. $(-\frac{8r}{7} + 9, \frac{8r}{7} + 27)$
- D. $(\frac{r}{3} + 9, -\frac{r}{3} + 27)$

Correct Answer: A

Rationale

Choice A is correct. Dividing both sides of the second equation in the given system by **3** yields $8x + 7y = 9$, which is the first equation in the given system. Therefore, the first and second equations represent the same line in the xy -plane. If the x - and y -coordinates of a point satisfy an equation, the point lies on the graph of the equation in the xy -plane. Choice A is a point with x -coordinate r and y -coordinate $-\frac{8r}{7} + \frac{9}{7}$. Substituting r for x and $-\frac{8r}{7} + \frac{9}{7}$ for y in the equation $8x + 7y = 9$ yields $8r + 7(-\frac{8r}{7} + \frac{9}{7}) = 9$. Applying the distributive property to the left-hand side of this equation yields $8r - 8r + 9 = 9$. Combining like terms on the left-hand side of this equation yields $9 = 9$, so the coordinates of the point $(r, -\frac{8r}{7} + \frac{9}{7})$ satisfy both equations in the given system. Therefore, for each real number r , the point $(r, -\frac{8r}{7} + \frac{9}{7})$ lies on the graph of each equation in the xy -plane for the given system.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.